

ED (EHSAN) SHAH HOSSEINI

Optical Network Development Engineer at AWS, US Citizen

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SUMMARY

Silicon photonics engineer with 20 years in CMOS-compatible photonic integration, spanning datacenter optics, consumer sensing, and defense applications. 53 granted US patents in optical phased arrays and LiDAR, and publications in Nature and Nature Communications (5,000+ citations). Principal Investigator on \$3.9M+ in DOE and DoD SBIRs at Analog Photonics, and co-author on the founding proposal for AIM Photonics, the \$600M+ Manufacturing USA institute. Currently building 800G/1.6T intrarack optical links at AWS; completed Andrew Ng's Deep Learning Specialization.

EXPERIENCE

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| 2026 | Amazon Web Services (AWS) , Optical Network Development Engineer, Cupertino, CA <ul style="list-style-type: none">· Design and validate 800G intrarack optical links (100G/lane, OSFP) deployed in AWS datacenter networks· Implement in-band 1PPS delivery of GPS-sourced timing from TOR to NIC, holding jitter under 10ns over existing high-speed optical links· Build an asynchronous side channel on the same optical link for telemetry and firmware updates, targeting 1-4 Mb/s alongside 1PPS delivery· Define architecture and vendor requirements for next-generation 1.6T links (200G/lane) |
| 20–26 | Apple Camera Technology Group , Silicon Photonics Lead, Cupertino, CA <ul style="list-style-type: none">· Coordinated cross-discipline team (VCSEL, TIA, driver, optics, applications) on R&D for SMI-based OCT consumer sensing; set system requirements (SNR, noise budget) and made component-level design decisions· Led R&D on coherent laser Doppler imaging for wearable health sensing· Built beam-scanning rig for 3D OCT capture and set up high-density SMI VCSEL array with coverglass backreflection compensation· Designed FMCW chirp pre- and post-processing to linearize all lasers across the SMI VCSEL array· Led MATLAB platform for end-to-end coherent sensing system simulation |
| 14–20 | Analog Photonics , Director of Advanced Projects and Photonics Integration <ul style="list-style-type: none">· Co-authored winning proposal for AIM Photonics, the \$600M+ Manufacturing USA institute for integrated photonics (\$110M AFRL, \$250M New York State, \$245M+ public/private match), 2015· Primary author of the first industry PDK at AIM Photonics, the foundry design kit used by external silicon-photonics customers; sustained \$1M+ annual AP funding through the institute· Principal Investigator on DOE SBIR program for photonic memory controller module (\$2.8M across Phases I, II, IIB): 30 Gb/s MZI-based athermal optical links and microring WDM toward 800GE on-board optics; multi-company consortium with Freedom Photonics and PLCC· Co-developed 400G-FR4 silicon photonics transceiver chipsets for CPO, OBO, and pluggable form factors (94 Gb/s/mm density, <10 pJ/bit, -5.4 dBm OMA at KP4 pre-FEC BER); demonstrated at OFC 2019· Invented two LiDAR architectures: WDM LiDAR combining multi-wavelength sources with optical phased array, and switched-array LiDAR using curved-focal-plane emitters with log₂(N) control signals; demonstrated chip-scale in silicon· Principal Investigator on five additional SBIRs (\$1M+ combined) across Army, Navy, and DoD: hybrid MOPA laser at 1550 nm (>1 W output, >1 GHz/μs chirp) with OPA-based coherent combining, UAV/tactical free-space optical comms, 3D optical sensors for Future Vertical Lift, and silicon photonics optogenetic transceivers |

- 11–14 | **MIT Photonics Microsystems Group**, Post-doctoral Associate
- Designed, fabricated, and demonstrated DFB and DBR erbium-doped lasers integrated into a CMOS-compatible silicon photonics fab, enabling on-chip light sources without III-V bonding
 - Developed germanium-on-silicon photodetectors from process up at 1550 nm: ~ 1 A/W responsivity, 50 GHz bandwidth, nA-scale dark current
 - Contributed to large-scale 2D nanophotonic optical phased array demonstration (Nature, 2013)
 - Contributed to ultralow-power athermal silicon modulator demonstration (Nature Communications, 2014)
 - Designed silicon photonics device library: PN-junction MZI modulators, microdisk and microring filters and modulators, edge couplers, and directional grating couplers

EDUCATION

- 03–11 | **Georgia Institute of Technology (Georgia Tech), Atlanta, GA**
 MS and PhD, Electrical Engineering (Photonics) | Minor: Physics
 Thesis: “High Quality Integrated Silicon Nitride Nanophotonic Structures for Visible Light Applications.”
- 99–03 | **Aryamehr (Sharif) University of Technology, Tehran, Iran**
 BSc, Electrical Engineering | Top 10 in EE, Emphasis: Fields & Waves in Communications

CONTINUING EDUCATION

- 23–25 | **DeepLearning.AI (Coursera)**
 Deep Learning Specialization (Andrew Ng)
 Machine Learning Specialization (Andrew Ng)

SELECTED PATENTS

Analog Photonics (2019–2024)

1. “Multi-static coherent LiDAR,” [US Patent 12,050,265 B2](#), 2024.
2. “Optical phased array light steering,” [US Patent 12,085,833 B2](#), 2024.
3. “Large scale steerable coherent optical switched arrays,” [Patent Application US20240329324A1](#), 2024.
4. “Aberration correction of optical phased arrays,” [US Patent 12,517,351 B2](#), 2026.
5. “Photonic die alignment,” [US Patent 11,693,196 B2](#), 2023.
6. “Photonics fabrication process performance improvement,” [US Patent 11,762,146 B2](#), 2023.
7. “Tunable optical structures,” [US Patent 11,269,236 B2](#), 2022.
8. “Optical switching using spatially distributed phase shifters,” [US Patent 11,422,431 B2](#), 2022.
9. “Speckle reduction in photonic phased arrays,” [US Patent 11,353,769 B2](#), 2022.
10. “Two dimensional optical phased arrays using edge-coupled integrated circuits,” [Patent Application WO2022051453A1](#), 2022.
11. “Array-based free-space optical communication links,” [US Patent 11,196,486 B2](#), 2021.
12. “Wavelength division multiplexed lidar,” [US Patent 11,061,140 B2](#), 2021.
13. “Optical phase shifter device,” [US Patent 10,809,591 B2](#), 2020.
14. “Large scale steerable coherent optical switched arrays,” [US Patent 10,338,321 B2](#), 2019.

Apple (2023–2024)

1. Depth sensing using multiple coherent emitters with interleaved VCSEL/detector arrays ([WO2024177716A1](#), 2024)

2. Coherent optical waveguide for wearable biometric sensing ([US20240057896A1](#), 2024)
3. “LiDAR array with vertically-coupled transceivers,” [Patent Application US20230366986A1](#), 2023.

MIT (2019)

1. “Photonic devices and methods of using and making photonic devices,” [US Patent 10,461,489 B2](#), 2019.
2. “Apparatus, systems, and methods for waveguide-coupled resonant photon detection,” [Patent Application US20170219776A1](#), 2017.

SELECTED PUBLICATIONS

Nature and High-Impact Journals

1. J. Sun, ..., E. S. Hosseini, “Large-scale nanophotonic phased array,” [Nature](#), 2013. (1,670 citations)
2. E. Timurdogan, ..., E. Shah Hosseini, “An ultralow power athermal silicon modulator,” [Nature Communications](#), 2014. (569 citations)

Silicon Photonics and Integrated Optics

1. E. Shah Hosseini, ..., “High quality planar silicon nitride microdisk resonators for integrated photonics in the visible wavelength range,” [Optics Express](#), 2009. (219 citations)
2. J. Sun, ..., E. S. Hosseini, “Large-scale silicon photonic circuits for optical phased arrays,” [IEEE JSTQE](#), 2013. (210 citations)
3. E. Shah Hosseini, ..., “Systematic design and fabrication of high-Q single-mode pulley-coupled planar silicon nitride microdisk resonators at visible wavelengths,” [Optics Express](#), 2010. (190 citations)
4. A. H. Atabaki, E. S. Hosseini, ..., “Optimization of metallic microheaters for high-speed reconfigurable silicon photonics,” [Optics Express](#), 2010. (158 citations)
5. J. Sun, E. S. Hosseini, ..., “Two-dimensional apodized silicon photonic phased arrays,” [Optics Letters](#), 2014. (128 citations)
6. J. D. B. Bradley, E. S. Hosseini, ..., “Monolithic erbium- and ytterbium-doped microring lasers on silicon chips,” [Optics Express](#), 2014. (117 citations)
7. Purnawirman, ..., E. S. Hosseini, “C- and L-band erbium-doped waveguide lasers with wafer-scale silicon nitride cavities,” [Optics Letters](#), 2013. (99 citations)
8. E. S. Hosseini, ..., “CMOS-compatible 75 mW erbium-doped distributed feedback laser,” [Optics Letters](#), 2014. (88 citations)
9. Purnawirman, ..., E. S. Hosseini, “Ultra-narrow-linewidth $\text{Al}_2\text{O}_3:\text{Er}^{3+}$ lasers with a wavelength-insensitive waveguide design on a wafer-scale silicon nitride platform,” [Optics Express](#), 2017. (56 citations)
10. Z. Su, ..., E. S. Hosseini, “Four-port integrated polarizing beam splitter,” [Optics Letters](#), 2014. (56 citations)

Foundry and Systems (2018)

1. E. Timurdogan, ..., E. S. Hosseini, “AIM process design kit (AIMPDKv2.0): Silicon photonics passive and active component libraries on a 300mm wafer,” [OFC](#), 2018. (54 citations)
2. C. V. Poulton, ..., E. Hosseini, “High-performance integrated optical phased arrays for chip-scale beam steering and LiDAR,” [CLEO](#), 2018. (50 citations)